

NT-C-029 | Linux Administration | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Systems & Cloud
Target learners	Learners targeting Linux, server and cloud-support roles.
Prerequisites	Basic computer and networking knowledge.
Approved duration	12-16 weeks / 140-180 guided hours.
Final outcome	Install, configure, secure, automate and troubleshoot Linux systems using documented administrative procedures.
Assessment profile	Infrastructure/Security

Approved tools / platforms

Linux distribution, Bash, systemd, SSH, firewall tools, storage tools, web service, Git.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.	Never recommend unsafe production testing or treat a scanner output as a verified finding without validation.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-029 | Linux Administration | NT-C-029-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Shell, paths, text tools and editors; Accounts, groups, ownership, ACL basics and sudo
Completion evidence	Command lab; Access lab
Section weight	20% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 1 - CLI and files

Objective: Shell, paths, text tools and editors

Observable evidence: Command lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Shell	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when shell is appropriate; apply it in a Linux Administration task; verify the result.	Command lab
paths	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when paths is appropriate; apply it in a Linux Administration task; verify the result.	Command lab
text tools	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when text tools is appropriate; apply it in a Linux Administration task; verify the result.	Command lab
editors	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when editors is appropriate; apply it in a Linux Administration task; verify the result.	Command lab

Module 2 - Users and permissions

Objective: Accounts, groups, ownership, ACL basics and sudo

Observable evidence: Access lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Accounts	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when accounts is appropriate; apply it in a Linux Administration task; verify the result.	Access lab
groups	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when groups is appropriate; apply it in a Linux Administration task; verify the result.	Access lab
ownership	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ownership is appropriate; apply it in a Linux Administration task; verify the result.	Access lab
ACL basics	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when acl basics is appropriate; apply it in a Linux Administration task; verify the result.	Access lab
sudo	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sudo is appropriate; apply it in a Linux Administration task; verify the result.	Access lab

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Command lab; Access lab, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-029 | Linux Administration | NT-C-029-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Software management, services and jobs; Partitions, filesystems, mounts and LVM basics
Completion evidence	Service lab; Storage exercise
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 3 - Packages and processes

Objective: Software management, services and jobs

Observable evidence: Service lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Software management	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when software management is appropriate; apply it in a Linux Administration task; verify the result.	Service lab
services	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when services is appropriate; apply it in a Linux Administration task; verify the result.	Service lab
jobs	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when jobs is appropriate; apply it in a Linux Administration task; verify the result.	Service lab

Module 4 - Storage

Objective: Partitions, filesystems, mounts and LVM basics

Observable evidence: Storage exercise

Concept	Approved knowledge statement	Application behaviour	Evidence
Partitions	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when partitions is appropriate; apply it in a Linux Administration task; verify the result.	Storage exercise
filesystems	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when filesystems is appropriate; apply it in a Linux Administration task; verify the result.	Storage exercise
mounts	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when mounts is appropriate; apply it in a Linux Administration task; verify the result.	Storage exercise
LVM basics	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when lvm basics is appropriate; apply it in a Linux Administration task; verify the result.	Storage exercise

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Service lab; Storage exercise, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-029 | Linux Administration | NT-C-029-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	IP, DNS, SSH, firewall and troubleshooting; systemd, web service, logs and rotation
Completion evidence	Network lab; Hosted service
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 5 - Networking

Objective: IP, DNS, SSH, firewall and troubleshooting

Observable evidence: Network lab

Concept	Approved knowledge statement	Application behaviour	Evidence
IP	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ip is appropriate; apply it in a Linux Administration task; verify the result.	Network lab
DNS	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dns is appropriate; apply it in a Linux Administration task; verify the result.	Network lab
SSH	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ssh is appropriate; apply it in a Linux Administration task; verify the result.	Network lab
firewall	A control that allows or blocks traffic according to defined rules. In Linux Administration, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when firewall is appropriate; apply it in a Linux Administration task; verify the result.	Network lab
troubleshooting	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when troubleshooting is appropriate; apply it in a Linux Administration task; verify the result.	Network lab

Module 6 - Services and logs

Objective: systemd, web service, logs and rotation

Observable evidence: Hosted service

Concept	Approved knowledge statement	Application behaviour	Evidence
systemd	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when systemd is appropriate; apply it in a Linux Administration task; verify the result.	Hosted service
web service	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when web service is appropriate; apply it in a Linux Administration task; verify the result.	Hosted service
logs	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logs is appropriate; apply it in a Linux Administration task; verify the result.	Hosted service
rotation	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when rotation is appropriate; apply it in a Linux Administration task; verify the result.	Hosted service

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Network lab; Hosted service, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-029 | Linux Administration | NT-C-029-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Variables, conditions, loops and scheduled jobs; Hardening, backup, monitoring and recovery
Completion evidence	Admin scripts; Managed Linux server
Section weight	30% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 7 - Shell scripting

Objective: Variables, conditions, loops and scheduled jobs

Observable evidence: Admin scripts

Concept	Approved knowledge statement	Application behaviour	Evidence
Variables	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when variables is appropriate; apply it in a Linux Administration task; verify the result.	Admin scripts
conditions	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when conditions is appropriate; apply it in a Linux Administration task; verify the result.	Admin scripts
loops	A design approach that organises behaviour and data around objects and clear responsibilities. In Linux Administration, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when loops is appropriate; apply it in a Linux Administration task; verify the result.	Admin scripts
scheduled jobs	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when scheduled jobs is appropriate; apply it in a Linux Administration task; verify the result.	Admin scripts

Module 8 - Security capstone

Objective: Hardening, backup, monitoring and recovery

Observable evidence: Managed Linux server

Concept	Approved knowledge statement	Application behaviour	Evidence
Hardening	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when hardening is appropriate; apply it in a Linux Administration task; verify the result.	Managed Linux server
backup	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when backup is appropriate; apply it in a Linux Administration task; verify the result.	Managed Linux server
monitoring	Collection and interpretation of metrics, logs and events to understand system health. In Linux Administration, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when monitoring is appropriate; apply it in a Linux Administration task; verify the result.	Managed Linux server
recovery	A defined competency within Linux Administration. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when recovery is appropriate; apply it in a Linux Administration task; verify the result.	Managed Linux server

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Admin scripts; Managed Linux server, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-029 | Linux Administration | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-029-SK01	CLI and files	Prerequisite foundation	Command lab	Critical
NT-C-029-SK02	Users and permissions	M01 plus earlier foundations	Access lab	Critical
NT-C-029-SK03	Packages and processes	M02 plus earlier foundations	Service lab	Supporting
NT-C-029-SK04	Storage	M03 plus earlier foundations	Storage exercise	Supporting
NT-C-029-SK05	Networking	M04 plus earlier foundations	Network lab	Supporting
NT-C-029-SK06	Services and logs	M05 plus earlier foundations	Hosted service	Critical
NT-C-029-SK07	Shell scripting	M06 plus earlier foundations	Admin scripts	Supporting
NT-C-029-SK08	Security capstone	M07 plus earlier foundations	Managed Linux server	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-029 | Linux Administration | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-029-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of CLI and files and Users and permissions?	The learner should provide Command lab; Access lab and explain how the work applies Shell, paths, text tools and editors; Accounts, groups, ownership, ACL basics and sudo. Recall alone is insufficient.
NT-C-029-S01-EX02	Scenario/remediation	A learner reports completing Users and permissions but cannot produce Access lab. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-029-S01, request Access lab, and allow a new equivalent test only after remediation.
NT-C-029-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Packages and processes and Storage?	The learner should provide Service lab; Storage exercise and explain how the work applies Software management, services and jobs; Partitions, filesystems, mounts and LVM basics. Recall alone is insufficient.
NT-C-029-S02-EX02	Scenario/remediation	A learner reports completing Storage but cannot produce Storage exercise. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-029-S02, request Storage exercise, and allow a new equivalent test only after remediation.
NT-C-029-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Networking and Services and logs?	The learner should provide Network lab; Hosted service and explain how the work applies IP, DNS, SSH, firewall and troubleshooting; systemd, web service, logs and rotation. Recall alone is insufficient.
NT-C-029-S03-EX02	Scenario/remediation	A learner reports completing Services and logs but cannot produce Hosted service. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-029-S03, request Hosted service, and allow a new equivalent test only after remediation.
NT-C-029-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Shell scripting and Security capstone?	The learner should provide Admin scripts; Managed Linux server and explain how the work applies Variables, conditions, loops and scheduled jobs; Hardening, backup, monitoring and recovery. Recall alone is insufficient.
NT-C-029-S04-EX02	Scenario/remediation	A learner reports completing Security capstone but cannot produce Managed Linux server. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-029-S04, request Managed Linux server, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Infrastructure/Security blueprint. Human-review every scored item before initial production release.

NT-C-029 | Linux Administration | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	User and backup automation	Permissions and shell	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Web server build	Service, network and logs	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Hardened Linux server	Monitoring, backup and recovery	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.

NT-C-029 | Linux Administration | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Linux Administrator Trainee	CLI 34% [critical]; services 33% [critical]; security 33%	Admin lab portfolio	If a critical skill is below 60 or admin lab portfolio is missing, return Improve First or Needs Evidence with the linked module/project.
2	System Support Engineer - Junior	Troubleshooting 50% [critical]; documentation 50% [critical]	Support runbook	If a critical skill is below 60 or support runbook is missing, return Improve First or Needs Evidence with the linked module/project.
3	Cloud/Linux Support Associate	Linux 34% [critical]; network 33% [critical]; automation 33%	Server capstone	If a critical skill is below 60 or server capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-029 | Linux Administration | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Linux Administration for?	Learners targeting Linux, server and cloud-support roles.
What should I know before starting?	Basic computer and networking knowledge.
How long is the approved learning journey?	12-16 weeks / 140-180 guided hours.
What will I be able to demonstrate?	Install, configure, secure, automate and troubleshoot Linux systems using documented administrative procedures.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Linux Administration knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-029; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```